

BeatEdit Symbolic Music Generation as Explicit Editing

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01 Why explicit editing?

Music creation is a process of revision.

Correct a wrong note

Change the error; retain the phrase.

Reshape accompaniment

Revise the left hand; preserve the melody.

Complete missing bars

Write new content into existing context.

Source and target overlap. The editor must locate what needs to change and preserve the rest.

02 The bottleneck is the representation: R1–R3

R1 · Atomic edit units

Each note should map to a compact, self-contained token group, allowing one edit label to address a complete musical target.

R2 · Source–target alignment

Source and target need stable positional correspondence, so edit labels can be extracted by directly comparing aligned units.

R3 · Edit locality

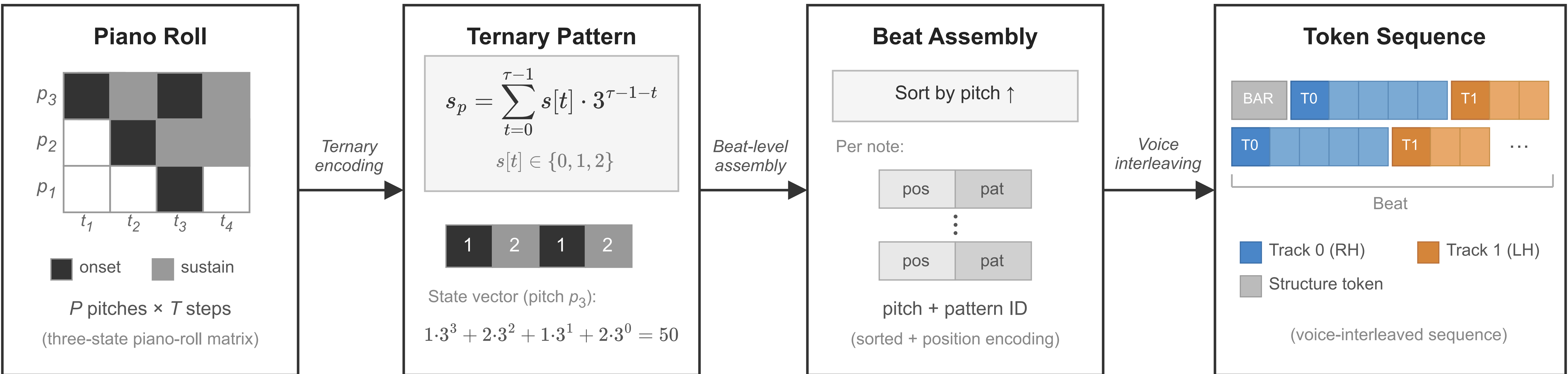
An edit should affect a bounded token span and preserve a valid encoding, without updating the rest of the sequence.

Encoding	R1	R2	R3	Bottleneck
MIDI-Like	×	×	×	Onset/offset decoupled
REMI	×	×	×	Insert/delete shifts positions
Structured	~	×	~	Relative time propagates
CPWord	~	×	~	No structural grid
BEAT (Rel.)	✓	✓	~	Within-beat pitch cascade
BEAT (Abs.)	✓	✓	✓	Meets all three requirements

✓ satisfied · × violated · ~ partial.

03 BEAT supplies a beat-aligned editing space

BEAT compresses within-beat rhythm into ternary patterns and aligns source and target on a fixed beat grid.



04 Choose the mechanism by edit density

SeqTag

SPARSE · ERROR CORRECTION

Detect erroneous tokens and assign edit tags, keeping low-confidence positions unchanged.

IterEdit

DENSE · ACCOMPANIMENT EDITING

Delete–insert–predict cycles revise accompaniment while preserving melody and structure.

TagFill

MISSING SPANS · SEGMENT COMPLETION

Allocate mask positions, then fill them in two confidence-ranked passes while preserving surrounding musical context.

05 Results: accuracy, latency, and encoding

Mechanism and encoding both matter: explicit editing improves accuracy and speed; edit-compatible encoding improves recovery. Correction/editing: beat exact match; completion: POP909 Note F1; encoding: recovery.

Correction

1.84×

SeqTag .726 vs AR-Detect .394
65 vs 9,677 ms

Editing

1.73×

IterEdit .480 vs AR-Detect .278
655 vs 9,458 ms

Completion

4.65×

TagFill-D .279 · Polyffusion .060
Note F1 · POP909 · 35 vs 1,747 ms

Encoding

4.81×

BEAT 48.1% vs REMI 10.0%
Recovery · Same SeqTag

Haoyu Gu · First author & presenter

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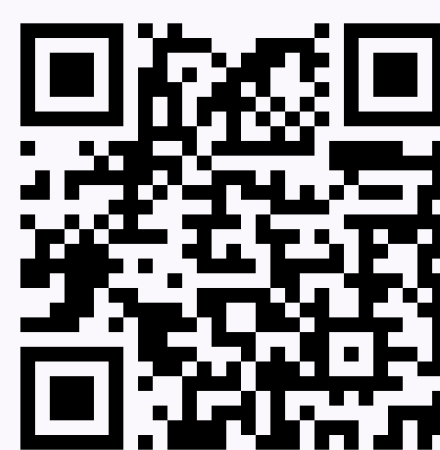
Research Interests: Symbolic Music Representation and Generation

haoyu-gu@outlook.com · Paper: arXiv:2607.11124 · Paper ID: 3781

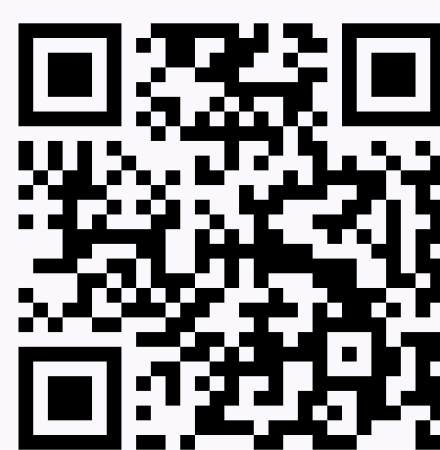
Previous Work: BEAT: Tokenizing and Generating Symbolic Music by Uniform Temporal Steps · ICML 2026



About Me



BEAT



Demo & Code



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